

OpenAI's math release: highlights and insights

What is in the **openai/math** repository (722 manuscripts produced by an unreleased internal OpenAI model), which claims are machine-checked, and how to read them, explained from first principles.

Manuscripts 722 grouped into result families	Result families 372 across 17 subjects	Problems posed to the model ~4,000 ≈3 h of thinking per result
Families with a Lean proof 127 34% of families; 162 papers (22%)	Lines of Lean code 25.9M in 121,734 files (Mathlib ≈ 2M)	Unfinished proofs ("sorry") 0 only standard axioms permitted

The short version

- **It is genuine.** The repository belongs to the OpenAI GitHub organisation, and your fork (oishik1306/math) matches it exactly.
- **The claims are extraordinary.** They include a zero-free half-plane for the Riemann zeta function, the Unique Games Conjecture, a matrix-multiplication exponent $\omega \leq 9/4$, the Mahler conjecture, the Birch–Swinnerton-Dyer formula in ranks 0 and 1, the Hodge conjecture for CM abelian varieties and the isomorphism of free group factors.
- **Trust must be judged result by result.** About a third of the families come with Lean proofs that a computer has checked against short, auditable statements. The rest are unrefereed papers, and several of the most famous claims are in that group.
- **The bottleneck is now human understanding.** Hundreds of manuscripts appeared within days, far faster than mathematicians can referee them.

How to use this guide: the contents page and the PDF bookmarks are clickable. In Section 4, each row of the summary table jumps to an explainer card. Appendix A lists all 372 families and links every title to its paper on GitHub. Badges show verification status:

☒ LEAN-CHECKED	a Lean proof of the headline statement is present	☐ PARTLY LEAN	only part of the family is formalised
☐ PAPER ONLY	no formal proof; an unrefereed paper		

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1 Is it really from OpenAI?

Yes. I checked the upstream repository directly and compared it with your fork:

Check	Result
Owner of github.com/openai/math	The OpenAI GitHub organisation
History and licence	A single commit, Apache-2.0 licence
Your fork, oishik1306/math	Identical README text and top-level layout (lean/, preprints/, reasoning_traces/, overview.pdf, CONTENTS.md), and the same single “Initial commit”
Internal metadata	overview.tex is dated 6 October 2026 with PDF author “OpenAI”; the Lean catalogue lists “OpenAI” as author
Press coverage	Several outlets report the release of 722 manuscripts in 372 families on 6 October 2026

Where this release fits

This is OpenAI’s third mathematics release in about ten weeks. The context helps when judging the claims.

Date	Event
Oct 2025	An OpenAI executive says GPT-5 “solved” 10 Erdős problems. The maintainer of the Erdős-problems website calls this a misrepresentation: the model had found existing papers. The episode is a reason to read claims carefully.
1 Aug 2026	Ten results with Lean certificates, headlined by the first explicit non-sofic group.
Aug 2026	arXiv 2608.14673, “A Human Audit of OpenAI’s AI-Generated Mathematical Proofs”, reviews those ten results. It finds no confirmed substantive error in a principal result, but notes uneven review depth and flags one chapter for major revision. It does <i>not</i> cover this repository.
8 Sep 2026	A 166-page Navier–Stokes finite-time blowup proof with Lean files. Mathematicians broadly accepted it because the Lean check passed, but called the paper very hard to read, and there was a public dispute about credit. That proof is <i>not</i> in this repository; family 376 here is a different Navier–Stokes result.
21 Sep 2026	Terence Tao announces an Advisory Group on Mathematics and AI. Its remit includes advising on how to coordinate a large release of AI-produced results.
6 Oct 2026	This repository is released. The same day, Tao announces “Hexagon”, a repository for research works prompted by the flood of LLM-assisted results. The README says OpenAI is “exploring community-hosted repositories”; whether that means Hexagon is my guess, not stated.

Note: arXiv, Tao’s blog and several news sites were blocked from my working environment, so details of the items above come from search-engine summaries of those pages.

2 The collection in numbers

According to the README, the model was posed about **4,000 open problems** as part of OpenAI's model evaluations, which were expanded after existing maths benchmarks “saturated”. Each result used on average about three hours of ChatGPT Pro-level thinking. OpenAI grouped the outputs into families (a main result plus companions, consequences or alternative proofs) and kept those judged significant. That gives 372 families and 722 manuscripts. Since families are not one-to-one with problems, 372 out of 4,000 is not a clean success rate.

Families by subject, and how many have a Lean proof

Subject	Families	Papers	Families with Lean	Share
Theoretical computer science	40	73	20	50%
Combinatorics	37	50	17	46%
Algebraic and complex geometry	36	89	6	17%
Number theory	31	58	6	19%
Probability and statistical mechanics	29	105	8	28%
Differential geometry	29	49	9	31%
Mathematical physics	25	59	6	24%
Operator algebras	19	31	6	32%
Algebra	18	29	7	39%
Topology	18	26	1	6%
Real and complex analysis	16	26	6	38%
Partial differential equations	16	29	3	19%
Convex and metric geometry	15	30	10	67%
Group theory	14	22	7	50%
Dynamical systems and ergodic theory	12	19	2	17%
Functional analysis	11	19	8	73%
Mathematical logic	6	8	5	83%
All subjects	372	722	127	34%

“Families with Lean” counts families where at least one paper has a formal proof of its main result. The paper-level figure is lower: 162 of 722 papers (22%).

Other facts worth knowing

- **Dates.** Manuscripts are dated between 10 September and 6 October 2026. 370 are dated 23–24 September alone and 112 are dated 5 October. These look like write-up dates rather than discovery dates.
- **Two flagship results came from a different process.** The README says the zero-free region for the Riemann zeta function and the Hodge conjecture for CM abelian varieties were exceptions to the standard procedure, and that the write-up of the 11/12 zero-free region was edited by humans. It does not say what the different procedure was.
- **Missing numbers.** Families are numbered up to 377, but 045, 061, 070, 123 and 163 do not exist. It looks as if five results were withdrawn before release; the repository does not say.
- **Reasoning summaries.** Abridged “chain of thought” summaries are included for 10 results, among them π , the Mahler conjectures, free group factors and Vlasov–Maxwell (see `reasoning_traces/`).
- **Versioning.** OpenAI says corrections will be released as new versions, and that earlier versions will remain available.

3 What “verified” means here

Everything in this guide depends on one distinction: a *claim* versus a *machine-checked claim*. Here is how that works from first principles.

What a proof assistant does

A normal maths paper is checked by people. A referee reads the argument and judges whether each step follows, which is slow and can miss errors in long proofs. A **proof assistant** such as **Lean** is a programming language in which definitions, theorem statements and proofs are written so precisely that a small, trusted program (the *kernel*) can check every logical step. If the kernel accepts a proof, the statement follows from the axioms, and no human judgement about the proof is needed. **Mathlib** is the shared Lean library of standard mathematics (real numbers, complex analysis, the zeta function and so on), roughly two million lines written by the community.

What still needs a human is the **statement**: does the formal theorem say what the paper claims? If a definition is subtly wrong, a perfect proof can prove the wrong thing.

Three layers of trust

Layer	What it is	Who checks it	Size
1. The paper	The human-readable PDF argument	Nobody yet; these are unrefereed preprints	722 papers
2. The Lean statement (“challenge”)	The theorem written using Mathlib’s standard definitions, with the proof left blank	Must be audited by a human against the paper	1 to about 250 lines each
3. The Lean proof (“solution”)	The complete formal proof	A computer: the Lean kernel plus the Comparator tool	25.9M lines in total

The tool **Comparator** compiles the proof and confirms that the proved theorem has *exactly* the statement in the challenge file. It also confirms that the proof uses only the three axioms behind all of Mathlib: propositional extensionality, quotient soundness and the axiom of choice. As a result, you do not have to trust 25.9 million lines. You only have to read the short statement.

Three examples of what you would actually audit

The Riemann zeta function has no zeros with real part above $7/8$ (family 003). This uses Mathlib’s own `riemannZeta`, so nothing was redefined:

```
theorem riemannZeta_ne_zero_of_seven_eighths_lt_re
  {s : ℂ} (hs : (7 / 8 : ℝ) < s.re) : riemannZeta s ≠ 0
```

The plane cannot be coloured with five colours (family 158). Any map from the plane (here $\mathbb{C}$) to five colours puts some pair of points at distance exactly 1 in the same colour:

```
def ProperColoring (colorCount : ℕ) (coloring : ℂ → Fin colorCount) : Prop :=
  ∀ point otherPoint : ℂ, ||point - otherPoint|| = 1 → coloring point ≠ coloring otherPoint

theorem no_proper_five_coloring :
  ¬ ∃ coloring : ℂ → Fin 5, ProperColoring 5 coloring
```

The symmetric Mahler conjecture (family 087). For every compact, convex, origin-symmetric K in $\mathbb{R}^n$ with nonempty interior, $\text{vol}(K) \cdot \text{vol}(K^\circ) \geq 4^n/n!$:

```
theorem symmetric_mahler {n : ℕ} (hn : 1 ≤ n)
  {K : Set (Fin n → ℝ)} (hK : IsCompact K) (hconv : Convex ℝ K)
  (hsym : ∀ x ∈ K, -x ∈ K) (hint : (interior K).Nonempty) :
  (4:ℝ)^n / (Nat.factorial n:ℝ) ≤
    (volume K).toReal * (volume (coordinatePolar K)).toReal
```

What I checked in the Lean code

- **All 405 Comparator challenges** permit only the three standard axioms; none allows anything extra.
- **Zero** occurrences of sorry (Lean's "proof omitted" placeholder) in 121,734 Lean files. A first search found 176 hits for "sorry|admit", but all were the English word "admit" in comments.
- **No** axiom declarations, and no native_decide (a shortcut that trusts compiled code).
- A dependency called iut (presumably Mochizuki's disputed inter-universal Teichmüller theory) is listed in the build file, but **nothing imports it**.
- **9 challenges define their own concepts** rather than using Mathlib's (for example k-server, Naimark's problem, Rokhlin mixing). The Unique Games statement runs to about 240 lines of encodings. Statements like these need the most careful human audit.

Caveats

- I did **not** compile the library; at 25.9 million lines it is far too large for this environment. In this guide, "Lean-checked" means a clean formal proof is *present*, not that I re-ran it.
- OpenAI's own catalogue (lean/formalization.yaml) says the formal proofs were written by AI agents ("automation: agent"), that the scope is "partial progress", and that review status is "unchecked".
- Often only some papers in a family are formalised, and the Lean statement can be narrower than the headline. For example, the zeta challenge covers ζ only, not every Dirichlet L-function.
- A formal proof establishes that a statement is true. It does not make the result understood.

4 Headline results

There are far too many families to cover individually (Appendix A lists them all), so this section picks the results a mathematician would recognise at once. Each card says what is claimed, explains the background from first principles, says why it matters, and states exactly what is formally verified. The wording of each claim follows OpenAI's catalogue.

At a glance

#	Claim (click to jump to its card)	Area	Status
003	$\zeta(s) \neq 0$ for $\text{Re } s > 7/8$ (quasi-Riemann hypothesis); no Landau–Siegel zeros	Number theory	✓ LEAN-CHECKED
017	The irrationality exponent of π is 2	Number theory	○ PAPER ONLY
002	Full BSD formula in Selmer corank 0 and 1; Goldfeld's conjecture	Number theory	○ PAPER ONLY
004	Hilbert's tenth problem over $\mathbb{Q}$ is undecidable	Number theory	○ PAPER ONLY
005	Catalan's constant is irrational	Number theory	○ PAPER ONLY
029	Artin's primitive root conjecture (infinitude), every base	Number theory	○ PAPER ONLY
030	Modularity of elliptic curves over imaginary quadratic fields	Number theory	○ PAPER ONLY
028	Gaussian moat: no bounded-step walk to infinity on Gaussian primes	Number theory	✓ LEAN-CHECKED
032	Hodge conjecture for CM abelian varieties	Algebraic geometry	○ PAPER ONLY
087	Mahler conjectures (symmetric and non-symmetric)	Convex geometry	✓ LEAN-CHECKED
102	The Unique Games Conjecture	Theoretical CS	✓ LEAN-CHECKED
107	Matrix multiplication exponent $\omega \leq 9/4$	Theoretical CS	✓ LEAN-CHECKED
103	L = RL = BPL (randomness does not help log-space)	Theoretical CS	○ PAPER ONLY
138	Subset Sum, integer multiplication and the DFT, all faster	Theoretical CS	🟡 PARTLY LEAN
158	The plane cannot be coloured with five colours	Combinatorics	✓ LEAN-CHECKED
156	Borsuk's conjecture fails in dimension 9	Combinatorics	✓ LEAN-CHECKED
155	Periodic tiling conjecture fails in dimension 3	Combinatorics	✓ LEAN-CHECKED
248	Thompson's group F is nonamenable	Group theory	✓ LEAN-CHECKED
196	Kaplansky's zero-divisor conjecture is false	Algebra	✓ LEAN-CHECKED
287	All free group factors are isomorphic	Operator algebras	○ PAPER ONLY
288	Kadison's similarity problem	Operator algebras	○ PAPER ONLY
246	Cannon's conjecture	Group theory	○ PAPER ONLY
304	Hilbert–Smith conjecture in every dimension	Topology	○ PAPER ONLY
362	Global smoothness for relativistic Vlasov–Maxwell	PDE	✓ LEAN-CHECKED

Number theory

FAMILY 003 · NUMBER THEORY · 2 OF 3 PAPERS FORMALISED

A zero-free half-plane for ζ : the quasi-Riemann hypothesis

 LEAN-CHECKED

The claim. Every Dirichlet L-function, including the Riemann zeta function $\zeta(s)$, has no zeros with real part greater than $7/8$. The same half-plane is zero-free for finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$. A companion paper reaches $\operatorname{Re} s > 11/12$ by a different route, and another rules out Landau–Siegel zeros.

From first principles. The zeta function is $\zeta(s) = 1 + 1/2^s + 1/3^s + \dots$. Euler noticed that it factors over the primes, $\zeta(s) = \prod_p 1/(1 - p^{-s})$, so ζ encodes every prime at once. Riemann showed that the zeros of ζ control how evenly the primes are spread. Concretely, if no zero has real part bigger than θ , then the number of primes up to x , written $\pi(x)$, differs from its smooth approximation $\operatorname{Li}(x) = \int_2^x dt / \log t$ by at most about x^θ . The Riemann Hypothesis says $\theta = 1/2$. Until now nobody could prove *any* θ below 1. The known zero-free regions hug the line $\operatorname{Re} s = 1$ and get thinner as you go up (their width shrinks roughly like $1/\log t$).

Why it matters. A fixed zero-free strip immediately gives $\pi(x) = \operatorname{Li}(x) + O(x^{7/8+\epsilon})$. That would be the first power-saving error term in the prime number theorem, 130 years after the theorem was proved. Separately, possible “exceptional” Siegel zeros near $s = 1$ have blocked effective bounds on class numbers and on primes in arithmetic progressions since the 1930s.

Verification. Two statements are formalised: the ζ statement (one line, shown in Section 3), and the Landau–Siegel statement. The second says there is an absolute $c > 0$ such that no real primitive Dirichlet character $\chi \bmod q$ has a real zero β with $1 - \beta < c / \log q$. The Dirichlet and Hecke versions and the 11/12 proof exist only as papers. The README says this result was produced outside the standard pipeline.

FAMILY 017 · NUMBER THEORY · 0 OF 1 PAPER FORMALISED

The irrationality exponent of π is exactly 2

 PAPER ONLY

The claim. For every $\epsilon > 0$ and every large enough q , every fraction p/q satisfies $|\pi - p/q| \geq q^{-2-\epsilon}$. As a consequence, the Flint–Hills series $\sum 1/(n^3 \sin^2 n)$ converges.

From first principles. Any irrational number x can be approximated by fractions; the question is how well, relative to the size of the denominator. Dirichlet showed that every irrational x has infinitely many fractions with $|x - p/q| < 1/q^2$; continued fractions produce them. The *irrationality exponent* $\mu(x)$ is the largest power μ for which $|x - p/q| < 1/q^\mu$ still has infinitely many solutions, so $\mu(x) \geq 2$ always. Almost every real number has $\mu = 2$ exactly, but proving this for a specific constant is notoriously hard. For π the best proven bound was $\mu(\pi) \leq 7.103$ (Zeilberger–Zudilin, 2020).

Why it matters. $\mu(\pi) = 2$ says that π is, in this precise sense, as hard to approximate as a “typical” number. The Flint–Hills series was a famous test case, since its convergence was known to follow from $\mu(\pi) < 5/2$.

Verification. Paper only. A reasoning summary is included; see insight 6 in Section 5.

FAMILY 002 + 006 · NUMBER THEORY · 0 OF 5 PAPERS FORMALISED

Birch–Swinnerton-Dyer in ranks 0 and 1, and Goldfeld’s conjecture

 PAPER ONLY

The claim. The full BSD leading-term formula, including finiteness of the Tate–Shafarevich group $\mathbb{W}$, holds for every elliptic curve over $\mathbb{Q}$ whose q -power Selmer group has corank 0 or 1 for some prime q . Goldfeld’s conjecture also holds: among the quadratic twists of any elliptic curve, ranks 0 and 1 each occur with density $1/2$. Together these give full BSD for 100% of quadratic twists.

From first principles. An elliptic curve is an equation such as $y^2 = x^3 + ax + b$. Its rational points form a finitely generated group (Mordell), and the *rank* r counts the independent points of infinite order. From counting solutions modulo each prime one also builds an analytic function, the L-function $L(E, s)$. The Birch–Swinnerton-Dyer conjecture (a Millennium Prize problem) says that r equals the order of vanishing of $L(E, s)$ at $s = 1$. It also predicts the exact leading coefficient in terms of arithmetic invariants, including the mysterious group $\mathbb{W}$. Gross–Zagier and

Kolyvagin (1980s) proved the rank equality when the analytic rank is 0 or 1, but the exact formula stayed open.

Why it matters. This is essentially the whole BSD conjecture in the ranks that occur most often, and Goldfeld's conjecture says those ranks are typical.

Verification. Paper only.

FAMILY 004 · NUMBER THEORY · 0 OF 2 PAPERS FORMALISED

Hilbert's tenth problem over $\mathbb{Q}$

☐ PAPER ONLY

The claim. No algorithm can decide whether a polynomial equation with integer coefficients has a solution in rational numbers.

From first principles. In 1900 Hilbert asked for a procedure to decide whether a polynomial equation (for example $x^3 + y^3 = 29z^3$) has a solution in whole numbers. Matiyasevich (1970), building on Davis, Putnam and Robinson, proved that no such algorithm exists: equations over the integers can encode arbitrary computer programs. Over the rationals the question stayed open for more than fifty years, because the tricks that force a variable to be an integer break down for fractions.

Why it matters. It settles one of the most-studied open problems between logic and number theory.

Verification. Paper only.

FAMILY 005 · NUMBER THEORY · 0 OF 1 PAPER FORMALISED

Catalan's constant is irrational

☐ PAPER ONLY

The claim. $G = 1 - 1/3^2 + 1/5^2 - 1/7^2 + \dots \approx 0.91597$ is not a fraction.

From first principles. Proving that a specific constant is irrational is often extremely hard. $\zeta(3)$ was only done by Apéry in 1978, and $\zeta(5)$ is still open. Catalan's constant is the alternating cousin of $\pi^2/8 = 1 + 1/3^2 + 1/5^2 + \dots$, and its irrationality has been a standard open problem for well over a century.

Verification. Paper only. The reasoning summary for the π result cites the determinant technique of this paper as an inspiration.

FAMILY 029 · NUMBER THEORY · 0 OF 2 PAPERS FORMALISED

Artin's primitive root conjecture (infinitude), for every base

☐ PAPER ONLY

The claim. Every integer a that is not -1 and not a perfect square is a primitive root modulo infinitely many primes. For each such a , at least $c_a x/(\log x)^2$ primes in $(x, 2x)$ qualify.

From first principles. Modulo a prime p , the nonzero remainders $1, \dots, p-1$ form a cyclic group under multiplication. A number a is a *primitive root* mod p if its powers $a, a^2, a^3, \dots$ run through all of them; for example, the powers of 3 modulo 7 are 3, 2, 6, 4, 5, 1. Artin (1927) conjectured that every a other than -1 and perfect squares is a primitive root for infinitely many primes. Hooley proved this in 1967 assuming a generalised Riemann Hypothesis. Unconditionally, Heath-Brown (1986) showed that at most two prime bases can fail, but could not say which, so not even $a = 2$ was known.

Verification. Paper only.

FAMILY 030 · NUMBER THEORY · 0 OF 1 PAPER FORMALISED

Modularity of elliptic curves over imaginary quadratic fields

☐ PAPER ONLY

The claim. Every elliptic curve over every imaginary quadratic field, such as $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-5})$, is modular, with matching local data at every place.

From first principles. The modularity theorem of Wiles, Taylor–Wiles and Breuil–Conrad–Diamond–Taylor says that every elliptic curve over $\mathbb{Q}$ comes from a modular form; this is what proved Fermat's Last Theorem. A central programme is to extend this to other number fields. Over imaginary quadratic fields the relevant automorphic

forms live on hyperbolic 3-space and do not come from algebraic geometry in the usual way, which makes the problem much harder. Only partial results were known.

Verification. Paper only.

FAMILY 028 · NUMBER THEORY · 1 OF 1 PAPER FORMALISED

The Gaussian moat conjecture

✓ LEAN-CHECKED

The claim. There is no infinite walk through distinct Gaussian primes with steps of bounded length. More strongly, for each step bound D , every cluster of Gaussian primes linked by steps of length at most D is finite, with a size bound that depends only on D .

From first principles. Gaussian integers are complex numbers $a + bi$ with a and b whole numbers, and Gaussian primes are those that cannot be factored further (for example 3, or $1 + 2i$). The question, posed in 1962, asks whether you can walk from the origin to infinity stepping only on Gaussian primes with steps no longer than some fixed length. Computer searches had found ever-wider prime-free “moats”, but there was no proof. On the ordinary number line the answer is easy, because prime gaps grow without bound.

Verification. Formalised.

Algebraic geometry

FAMILY 032 · ALGEBRAIC GEOMETRY · 0 OF 8 PAPERS FORMALISED

The Hodge conjecture for CM abelian varieties

○ PAPER ONLY

The claim. The rational Hodge conjecture holds for every complex abelian variety with complex multiplication, in every dimension and codimension. Companion papers prove it for arbitrary products of K3 surfaces. Via Milne's theorems this gives the Tate conjecture for all abelian varieties over finite fields.

From first principles. A smooth complex projective variety is a shape cut out by polynomial equations. Its cohomology (a way of counting holes of each dimension) contains special classes called *Hodge classes*, which look algebraic from the point of view of analysis. Every subvariety produces a Hodge class. The Hodge conjecture, a Millennium Prize problem, says that with rational coefficients every Hodge class arises this way. Abelian varieties are higher-dimensional analogues of elliptic curves, and those with complex multiplication (CM) have extra symmetry that creates many “exotic” Hodge classes. They have long been a test case where the conjecture was open.

Why it matters. This would be one of the largest known cases of a Millennium problem, with consequences in characteristic p as well.

Verification. Paper only. The README says this result was produced outside the standard pipeline.

Convex geometry

FAMILY 087 · CONVEX GEOMETRY · 2 OF 3 PAPERS FORMALISED

The Mahler conjectures

✓ LEAN-CHECKED

The claim. For every convex body K in $\mathbb{R}^n$ that is symmetric about the origin, $\text{vol}(K) \cdot \text{vol}(K^\circ) \geq 4^n/n!$. Equality holds exactly for Hanner polytopes, such as the cube. The non-symmetric version holds too, with simplices as the minimisers.

From first principles. A convex body is a bounded convex shape with interior, like a ball, a cube or an ellipsoid. Its *polar* is $K^\circ = \{y : x \cdot y \leq 1 \text{ for every } x \in K\}$. Stretching K squashes K° and vice versa, so the product $\text{vol}(K) \cdot \text{vol}(K^\circ)$ does not change under any invertible linear map; it measures shape alone. The ball maximises it (the Blaschke–Santaló inequality). Mahler (1939) conjectured that the cube minimises it among symmetric bodies, which gives the bound $4^n/n!$. This was known only in dimensions 2 and 3, and dimension 3 was settled only in

2020.

Why it matters. It is a basic inequality of convex geometry, with links to the geometry of numbers, functional analysis and symplectic geometry. The family also computes the Gromov width of $K \times K^\circ$.

Verification. The symmetric inequality is formalised for every n ; the statement is in Section 3. The non-symmetric case exists only as a paper. A reasoning summary is included.

Theoretical computer science

FAMILY 102 · THEORETICAL CS · 3 OF 5 PAPERS FORMALISED

The Unique Games Conjecture

✓ LEAN-CHECKED

The claim. Khot's Unique Games Conjecture is true. Separate direct proofs show it is NP-hard to beat the Goemans–Williamson ratio for Max-Cut, and to approximate Vertex Cover below a factor of 2.

From first principles. Many optimisation problems are NP-hard to solve exactly, so we ask how close an efficient algorithm can get. A *Unique Games* instance is a graph, a set of k labels, and on each edge a permutation π saying “if this end gets label a , the other end must get $\pi(a)$ ”. If every constraint can be satisfied, a solution is easy to find: label one vertex and propagate. Khot (2002) conjectured that the approximate version is hard. For any $\epsilon, \delta > 0$, it should be NP-hard to tell apart instances where at least $1 - \epsilon$ of the constraints can be satisfied from instances where at most δ can.

Why it matters. If true, many known algorithms are exactly optimal (unless $P = NP$): the 0.878 ratio of Goemans–Williamson for Max-Cut, factor 2 for Vertex Cover, and, by Raghavendra's theorem, the basic semidefinite-programming algorithm for every constraint satisfaction problem. It is arguably the central open problem in hardness of approximation.

Verification. The conjecture itself is formalised, as a polynomial-time Turing-machine reduction from 3-SAT to Unique Games with completeness at least $1 - \epsilon$ and soundness at most δ , for all $0 < \epsilon, \delta < 1/2$. The Max-Cut and Vertex Cover theorems are formalised too. The statement uses about 240 lines of definitions, so it deserves close human audit. A reasoning summary covers a related paper.

FAMILY 107 · THEORETICAL CS · 3 OF 3 PAPERS FORMALISED

Matrix multiplication with exponent $\omega \leq 9/4$

✓ LEAN-CHECKED

The claim. Two $n \times n$ complex matrices can be multiplied with $O(n^{9/4+\epsilon})$ arithmetic operations, so $\omega \leq 2.25$. The dual exponent satisfies $\alpha > 0.465$. Over every field $\omega < 2.37106$, and outside finitely many positive characteristics $\omega < 2.258$.

From first principles. Multiplying two $n \times n$ matrices the schoolbook way takes about n^3 multiplications: each of the n^2 entries of the answer is a sum of n products. In 1969 Strassen found how to multiply 2×2 matrices with 7 multiplications instead of 8; applied recursively, this gives $n^{2.807}$. The exponent ω is the best achievable power, and $\omega \geq 2$ because the answer alone has n^2 entries. Decades of increasingly elaborate methods lowered the bound to about 2.371 (2024). Between 1990 and 2024 it moved by only about 0.004.

Why it matters. Matrix multiplication sits underneath linear algebra, graph algorithms and much of complexity theory, so ω appears in the running time of hundreds of algorithms. A drop from 2.371 to 2.25 would be by far the largest improvement in more than 35 years.

Verification. All three bounds are formalised. The challenge defines ω in the textbook way: straight-line arithmetic programs over $\mathbb{C}$, where each $+$, $-$ and $\times$ costs one step.

FAMILY 103 · THEORETICAL CS · 0 OF 1 PAPER FORMALISED

Derandomising logarithmic space: $L = RL = BPL$  PAPER ONLY

The claim. Every problem a randomised algorithm can solve with $O(\log n)$ memory, with one- or two-sided bounded error, can be solved deterministically with $O(\log n)$ memory. The paper gives an explicit compiler.

From first principles. The resource here is memory rather than time. L is the class of problems a deterministic machine can solve with memory proportional to $\log n$, which is enough for a few pointers into the input. RL and BPL let the machine flip coins, with one-sided or two-sided error. Does randomness ever help? For space, the evidence said no, but the best unconditional result for decades was $BPL \subseteq SPACE(\log^{3/2} n)$ (Saks–Zhou, 1999). Reingold's 2005 proof that undirected connectivity is in L was a famous special case.

Verification. Paper only.

FAMILY 138 + 109 + 130 · THEORETICAL CS · 1 OF 5 PAPERS FORMALISED

Faster basic algorithms: Subset Sum, integer multiplication, Fourier transform PARTLY LEAN

The claim. Three classic running-time barriers are broken. Subset Sum runs in $O(2^{0.49n})$ time (family 138). Integers multiply in $O(n (\log n)^{1-\kappa})$ time with $\kappa = 2^{-182}$ (family 109). The discrete Fourier transform uses $O(n (\log n)^{1-\delta})$ operations with $\delta = 10^{-13}$ (family 130).

From first principles. *Subset Sum* asks whether some of n given integers add up to a target. Trying all subsets takes 2^n steps. “Meet in the middle” (Horowitz–Sahni, 1974) cuts this to about $2^{n/2}$, and that exponent had not been beaten for worst-case inputs in 50 years. For *integer multiplication*, Harvey and van der Hoeven (2019) reached $n \log n$, long conjectured to be optimal; the new bound beats it on multitape Turing machines, although a gain of 2^{-182} in the exponent of a logarithm is invisible in practice. The *fast Fourier transform* has run in $n \log n$ since 1965; the claim beats that in an exact-arithmetic model.

Verification. Subset Sum and integer multiplication are paper only. For the Fourier transform, only the companion paper “Finite tensor savings and exact Fourier circuits” is formalised, not the main one.

Combinatorics

FAMILY 158 · COMBINATORICS · 1 OF 1 PAPER FORMALISED

The plane cannot be coloured with five colours LEAN-CHECKED

The claim. Every colouring of the plane with five colours has two points at distance exactly 1 with the same colour. There are no restrictions on what the colour classes look like.

From first principles. Colour every point of the plane so that any two points exactly one unit apart get different colours. What is the smallest number of colours that works? This is the “chromatic number of the plane”, from the Hadwiger–Nelson problem (1950). A tiling by hexagons of diameter slightly under 1 shows that 7 colours suffice. Lower bounds come from finite unit-distance graphs that cannot be coloured with few colours. For decades the best lower bound was 4, until de Grey (2018) found a graph that needs 5.

Why it matters. The answer is now pinned to 6 or 7.

Verification. Formalised. The Lean statement is three lines; see Section 3.

FAMILY 156 · COMBINATORICS · 1 OF 1 PAPER FORMALISED

Borsuk's conjecture fails in dimension 9 LEAN-CHECKED

The claim. There is a compact set in $\mathbb{R}^9$ that cannot be split into 10 pieces of smaller diameter.

From first principles. The diameter of a set is the largest distance between two of its points. Borsuk (1933) asked whether every bounded set in $\mathbb{R}^n$ can be split into $n + 1$ pieces, each of strictly smaller diameter; a ball

shows that $n + 1$ cannot be lowered. The answer is yes in dimensions 1, 2 and 3. Kahn and Kalai (1993) showed it fails in dimension 1325, and the smallest dimension with a known counterexample was 64 (Jenrich–Brouwer, 2014).

Why it matters. The unknown range shrinks to dimensions 4 to 8. The example is built from rank-one orthogonal projectors onto lines in $\mathbb{R}^4$.

Verification. Formalised.

FAMILY 155 · COMBINATORICS · 1 OF 1 PAPER FORMALISED

The periodic tiling conjecture fails in dimension 3

✓ LEAN-CHECKED

The claim. There is a finite set of cells in $\square^3$ whose translates tile space, but never periodically. Thickening the cells to unit cubes gives the same result in $\mathbb{R}^3$.

From first principles. Suppose copies of a single shape, moved only by translations (no rotations), can fill space without overlaps. Must there then also be a *periodic* way to do it, one that repeats in every direction? This is true in dimension 1, and for the lattice $\square^2$ (Bhattacharya, 2016). Greenfeld and Tao (2022) disproved it in some very high dimension.

Why it matters. Since the conjecture holds in $\square^2$, dimension 3 is the smallest possible counterexample.

Verification. Formalised.

Groups, algebra and operator algebras

FAMILY 248 · GROUP THEORY · 1 OF 1 PAPER FORMALISED

Thompson's group F is nonamenable

✓ LEAN-CHECKED

The claim. Thompson's group F is not amenable.

From first principles. A group G is *amenable* if every bounded function on G can be given an “average” that does not change when the function is shifted by group elements. Finite groups and abelian groups are amenable; the free group on two generators is not. Thompson's group F consists of the piecewise-linear bijections of $[0, 1]$ whose breakpoints are dyadic rationals and whose slopes are powers of 2. It sits right on the boundary: it contains no free subgroup, the usual obstruction, yet no averaging had ever been found.

Why it matters. The question has been asked since about 1979 and is among the best-known problems in group theory. Several claimed proofs, in both directions, were later found to be wrong.

Verification. Formalised. The challenge defines F concretely and states that no invariant mean exists.

FAMILY 196 · ALGEBRA · 1 OF 1 PAPER FORMALISED

A counterexample to Kaplansky's zero-divisor conjecture

✓ LEAN-CHECKED

The claim. There is a finitely presented torsion-free group G whose group algebra $F_2[G]$ has nonzero zero divisors.

From first principles. The group algebra of G consists of finite formal combinations of group elements, multiplied using the group law. If G has an element g of finite order n, then $(1 - g)(1 + g + \dots + g^{n-1}) = 0$, so zero divisors appear. Kaplansky (1950s) conjectured that torsion-free groups never have them. Gardam (2021) disproved the related unit conjecture; the zero-divisor conjecture remained open.

Verification. Formalised.

FAMILY 287 · OPERATOR ALGEBRAS · 0 OF 1 PAPER FORMALISED

All free group factors are isomorphic☐ PAPER ONLY

The claim. $L(F_2) \cong L(F_3)$. Hence all free group factors $L(F_n)$, $n \geq 2$, and the interpolated ones, including $L(F_\infty)$, are isomorphic.

From first principles. From any group G one builds a von Neumann algebra $L(G)$: the closure of the algebra of operators that G generates when it acts on $\ell^2(G)$. For the free groups F_n on n generators these algebras are “factors” with a natural trace. Whether $L(F_2)$ and $L(F_3)$ are isomorphic is one of the most famous problems in operator algebras. In the 1990s Dykema and Rădulescu showed that either all $L(F_n)$ with $n \geq 2$ are isomorphic, or all are pairwise different.

Why it matters. This settles the dichotomy in the “all isomorphic” direction. Voiculescu’s free probability theory was developed largely as a tool for this question.

Verification. Paper only. A reasoning summary is included.

FAMILY 288 · OPERATOR ALGEBRAS · 0 OF 1 PAPER FORMALISED

Kadison’s similarity problem☐ PAPER ONLY

The claim. Every bounded unital homomorphism from a C^* -algebra into the bounded operators on a Hilbert space becomes a $*$ -homomorphism after conjugation by a bounded invertible operator.

From first principles. A C^* -algebra is an algebra of operators closed under taking adjoints and limits. Kadison (1955) asked whether representations that are merely bounded are always, after a change of coordinates, of the well-behaved “self-adjoint” kind. This was known for important classes of algebras but open in general.

Verification. Paper only.

Topology, geometry and PDE

FAMILY 246 · GROUP THEORY · 0 OF 1 PAPER FORMALISED

Cannon’s conjecture☐ PAPER ONLY

The claim. Every hyperbolic group whose boundary is a 2-sphere acts geometrically on hyperbolic 3-space. The torsion-free ones are fundamental groups of closed hyperbolic 3-manifolds.

From first principles. A hyperbolic group is one whose Cayley graph looks negatively curved from far away, and it has a boundary “at infinity”. Cannon’s conjecture (1980s) says that if this boundary is a 2-sphere, the group really comes from three-dimensional hyperbolic geometry. It is a key missing piece in recognising 3-manifold groups from their large-scale geometry.

Verification. Paper only.

FAMILY 304 · TOPOLOGY · 0 OF 1 PAPER FORMALISED

The Hilbert–Smith conjecture in every dimension☐ PAPER ONLY

The claim. Every locally compact group acting faithfully and continuously on a connected finite-dimensional manifold is a Lie group.

From first principles. Hilbert’s fifth problem asked whether continuous groups of transformations are automatically smooth (Lie groups). The main case was solved by Gleason, Montgomery and Zippin in the 1950s. The Hilbert–Smith conjecture is the remaining general form, and its crucial case is to rule out actions of the p -adic integers $\mathbb{Z}_p$. It was known only for manifolds of dimension at most 3 (Pardon, 2013).

Verification. Paper only.

FAMILY 362 · PDE · 1 OF 1 PAPER FORMALISED

Global smoothness for the relativistic Vlasov–Maxwell system

 LEAN-CHECKED

The claim. In three dimensions, take any smooth initial data with compactly supported particle density and finite-energy fields. The solution then exists, is unique and stays smooth for all time.

From first principles. This system models a collisionless plasma: a cloud of charged particles moving near the speed of light, described by a density $f(t, x, p)$ over position and momentum, coupled to the electric and magnetic fields the particles themselves generate (Maxwell's equations). The danger is that particles concentrate and the fields blow up. Glassey and Strauss (1986) proved global smoothness provided particle momenta stay bounded. Removing that condition was a long-standing open problem in kinetic theory.

Verification. Formalised. A reasoning summary is included.

5 Key insights

1. Trust has to be judged result by result

The repository mixes machine-checked theorems with unrefereed papers, often within the same family. Of the headline claims in Section 4, these are split as follows:

☒ LEAN-CHECKED	☐ PAPER ONLY
ζ zero-free for $\text{Re } s > 7/8$; no Siegel zeros	BSD formula in ranks 0/1; Goldfeld
Unique Games Conjecture	Hodge for CM abelian varieties
$\omega \leq 9/4$	Modularity over imaginary quadratic fields
Symmetric Mahler conjecture	Hilbert's tenth problem over $\mathbb{Q}$
Thompson's group F nonamenable	Irrationality exponent of π
Plane not 5-colourable	Catalan's constant irrational
Borsuk fails in dimension 9	Artin's conjecture
Periodic tiling fails in dimension 3	$L = RL = BPL$
Kaplansky zero-divisor counterexample	Subset Sum, integer multiplication
Gaussian moat	Free group factors isomorphic
Relativistic Vlasov–Maxwell	Kadison similarity
	Cannon's conjecture
	Hilbert–Smith conjecture

Read the left column as “true, provided the formal statement says what the paper says”. Read the right column as serious but unrefereed preprints from an author with a strong, though imperfect, track record. Many of the most famous claims are in the right column.

2. A formal proof turns refereeing into statement-checking

This is the real change in how mathematics gets verified. Nobody can referee 25.9 million lines, but anyone with some Lean experience can read a one-line zeta statement or a three-line colouring statement and decide whether it matches the paper. Three risks remain. Definitions can be subtly mismatched (9 challenges define their own concepts). Often only part of a family is formalised. And OpenAI marks its own formalisations as “unchecked”.

3. Formal coverage is thinnest where the most famous claims are

The share of families with a Lean proof is high in mathematical logic (83%), functional analysis (73%), convex geometry (67%), theoretical CS and group theory (50% each). It is low in number theory (19%), PDE (19%), algebraic geometry (17%) and topology (6%). The likely reason is Mathlib itself. It has deep foundations for analysis, combinatorics and algebra, but little of the machinery that modern number theory and algebraic geometry rely on (for example étale cohomology, automorphic forms and Hodge theory). Formalising BSD or Hodge would first require building those theories.

4. Disproofs are easier to certify than proofs

About 64 families are counterexamples or negative results. Of these, 34 (53%) have a Lean proof, compared with 93 of 308 (30%) positive results. This is a keyword-based estimate. The pattern makes sense: a counterexample is a concrete object that a computer can check, whereas a positive theorem usually needs a whole supporting theory.

5. The model builds on its own results, so errors can be correlated

Results cite each other. The π reasoning trace explicitly draws on “OpenAI’s Catalan determinants” (family 005). The BSD result (002) combines with Goldfeld (006) to reach 100% of twists. Milne’s rationality conjecture (001) uses the Hodge result (032). This compounding is powerful, but it also links failures: a flaw in one unformalised base result would spread to everything built on it.

6. Inside a reasoning summary: how the π result was found

The π summary runs to 42 pages and 38 sections, most of them dead ends: Padé approximants, modular forms, elliptic periods, BBP formulas, theta functions and more, each abandoned when a height or denominator cost could not be paid. The model was initially aiming only for $\mu(\pi) < 5/2$, enough for Flint–Hills, and found a candidate bound of $62/25 = 2.48$. Then, while “stress-testing ... its unexpectedly strong exponent”, it noticed that its method could be pushed all the way down to 2. The summary also shows repeated self-audits, with the model looking for counterexamples to its own lemmas. Note that these are abridged summaries, not full transcripts.

7. The flagship results came from a different, undisclosed process

The README’s description of the standard procedure (about 3 hours of compute per result) explicitly excludes the zeta zero-free region and the Hodge result for CM abelian varieties, and says the 11/12 write-up was edited by humans. How those results were produced, and with how much compute or human involvement, is not stated. That matters for judging what the “standard” model can do.

8. Production has outrun review

With 370 manuscripts dated within two days, normal refereeing cannot keep up. Several signals point to a field adjusting in real time:

- readability complaints about the earlier Navier–Stokes paper, despite its Lean check;
- credit disputes;
- Tao’s advisory group, and the new Hexagon repository launched on release day;
- five family numbers that silently disappeared;
- the October 2025 episode in which claimed “solutions” turned out to be literature lookups.

Expect corrections. OpenAI says it will publish them as new versions.

A practical checklist for any claim in this repository

- Find the family in Appendix A and look at the Lean column.
- If it has a Lean proof: open `lean/formalization.yaml`, find the challenge file under `lean/ComparatorChallenges/`, read the statement and compare it with the paper’s abstract. Note any custom definitions.
- If not: treat it as an unrefereed preprint. Check whether its catalogue entry says it depends on other families (“Together with result ...”), and how those are verified.
- Look for independent follow-up, such as expert commentary, reuse in other papers, Hexagon, or arXiv audits like the one done for the August results.

6 What this review did and did not do

Did: confirmed provenance on GitHub; parsed all 372 catalogue entries; mapped all 162 formalised papers to their families; scanned every Lean file for `sorry`, axioms and `native_decide`; read all 405 Comparator configurations and about eight challenge statements in detail (zeta, Landau–Siegel, Mahler, five-colouring, matrix multiplication, Thompson’s group, Unique Games, Vlasov–Maxwell); and read the π reasoning summary.

Did not: compile the Lean library; check any paper's mathematics; or read most of the 722 manuscripts. Background facts about earlier work (previous records and dates) are from my own knowledge and are worth double-checking before you rely on them. Outside reactions come from search-engine summaries, because several sites were blocked from my environment.

Appendix A All 372 result families

Every family in OpenAI's catalogue, grouped by subject in catalogue order. Titles link to the first paper of each family on GitHub. "Lean" shows how many of the family's papers have a formal proof. A "guide" link jumps to that family's card in Section 4.

Number theory · 31 families, 6 with Lean

#	Result	Papers	Lean
001	Milne's rationality conjecture	1	—
002	The full Birch–Swinnerton-Dyer formula in Selmer coranks zero and one ▶ guide	3	—
003	The quasi-Riemann hypothesis ▶ guide	3	● 2 of 3
004	Hilbert's tenth problem over $\mathbb{Q}$ ▶ guide	2	—
005	Irrationality of Catalan's constant ▶ guide	1	—
006	Goldfeld's conjecture ▶ guide	2	—
007	Two-point Chowla and the corrected binary Elliott conjecture	1	—
008	The Deligne–Drinfeld conjecture	1	● 1 of 1
009	Bogomolov–Pop and Milnor K-theoretic reconstruction	3	—
010	Fontaine–Mazur modularity at the prime 2 and 2-adic pro-modularity	3	—
011	The Ford–Konyagin–Luca conjecture on prime predecessors	3	—
012	Independent largest prime factors of consecutive integers	1	—
013	Ostmann's inverse Goldbach conjecture	1	—
014	Restricted geometric Langlands, generic Ramanujan and Arthur parameters	8	—
015	Torus-packet equidistribution in prime, quartic, and sextic degrees	3	—
016	Zilber–Pink in abelian varieties and the Siegel threefold	4	—
017	The irrationality exponent of π is 2 ▶ guide	1	—
018	The Margulis–Platonov conjecture over global fields	2	—
019	The p-adic section conjecture	2	—
020	Squarefree quartics and power-free polynomial values	1	● 1 of 1
021	A quadratic bound for Jacobsthal's function	1	● 1 of 1
022	The weak inhomogeneous Duffin–Schaeffer conjecture	1	—
023	Patterson's first moment for cubic Gauss sums	1	—
024	An asymptotic formula for the number of totients	1	—
025	Erdős's short Egyptian-fraction conjecture	1	● 1 of 1
026	Positive lower density of large prime gaps	1	—
027	Integral density on curve character varieties	1	—
028	The Gaussian moat conjecture ▶ guide	1	● 1 of 1
029	Artin's primitive root conjecture: infinitude for every base ▶ guide	2	—
030	Modularity over imaginary quadratic fields ▶ guide	1	—
031	Uchida's conjecture for open Galois homomorphisms	1	—

Algebraic and complex geometry · 36 families, 6 with Lean

#	Result	Papers	Lean
032	The rational Hodge conjecture for CM abelian varieties and products of K3 surfaces ► guide	8	—
033	Campana's orbifold litaka conjecture and logarithmic subadditivity	5	—
034	Log abundance and effective litaka fibrations	14	—
035	Threefold log abundance in numerical dimension one in characteristic $p > 3$	2	—
036	Numerical semiampleness and generalized minimal models	4	—
037	The sharp ordinary-double-point volume gap	2	—
038	Fujita's freeness conjecture	1	—
039	Nagata's conjecture and maximal Seshadri constants	4	● 2 of 4
040	Bloch's conjecture for complex surfaces	1	—
041	Hyperkähler SYZ and projective-space bases	2	—
042	Oka classification for K3 surfaces and other compact complex surfaces	1	—
043	P = W for fixed-determinant moduli spaces	1	—
044	The equivariant cohomological Hikita conjecture for quivers	1	—
046	Counterexamples to Shafarevich holomorphic convexity	2	—
047	Complex counterexamples to cancellation and affine fibrations	1	● 1 of 1
048	A characteristic-zero counterexample to Lipman–Zariski	1	—
049	A stable-coordinate counterexample in four variables	2	● 1 of 2
050	A counterexample to Griffiths' positivity conjecture	1	● 1 of 1
051	Kobayashi's canonical-ampleness conjecture	1	—
052	Tangent-bundle splittings and universal covers	2	● 1 of 2
053	A counterexample to Pixton's original completeness conjecture	1	—
054	Counterexamples to Kuznetsov's rationality conjecture	1	—
055	Toda's Gepner conjecture and large-volume stability	2	—
056	Termination of fourfold minimal model programs	5	—
057	Campana's abelianity conjecture and special varieties	3	—
058	The Kollár–Pardon universal-cover conjecture	2	● 1 of 2
059	Counterexamples to Zariski's multiplicity conjecture	2	—
060	The Global Spherical Shell conjecture	1	—
062	The LeBrun–Salamon conjecture and projective contact classification	1	—
063	The generalized Mukai conjecture	1	—
064	The μ-constant problem for surface singularities	1	—
065	Virasoro constraints for complete intersections and projective bundles	2	—
066	Bounded klt complements for Fano contractions	2	—
067	The Campana–Petersen conjecture in dimension six	1	—
068	Anticanonical nonvanishing under smooth semipositivity	7	—
069	Quantum geometric Langlands at irrational level	1	—

Real and complex analysis · 16 families, 6 with Lean

#	Result	Papers	Lean
071	Koebe's circle-domain conjecture and circle-domain rigidity	2	—
072	Brennan's conjecture and a counterexample to Kraetzer's prediction	2	● 2 of 2
073	The Falconer distance conjecture	1	● 1 of 1
074	Takeya in three and four dimensions	2	—
075	The $L \log L$ Fourier-convergence conjecture	1	—
076	Ultraflat real Littlewood polynomials	3	● 1 of 3
077	Fourier restriction for positively curved surfaces	2	—
078	The three-dimensional Bochner–Riesz conjecture	1	—
079	Sogge's local smoothing conjecture in dimension three	1	—
080	The Sobolev endpoint in Carleson's Schrödinger convergence problem	2	—
081	The David–Semmes Riesz-transform problem in higher codimension	1	● 1 of 1
082	Maximal and variational bounds for the triangular Hilbert transform	3	● 1 of 3
083	Stein's conjecture for Hilbert transforms along Lipschitz directions	1	—
084	The geometric case of the Erdős similarity conjecture	2	—
085	Endpoint regularity of the planar centered maximal function	1	● 1 of 1
086	An L^3 bound for the trilinear Hilbert transform	1	—

Convex and metric geometry · 15 families, 10 with Lean

#	Result	Papers	Lean
087	The Mahler conjectures and symplectic width ► guide	3	● 2 of 3
088	Petty's projection-volume conjecture and simplex counterexamples	2	● 2 of 2
089	Bounded-distortion L_1 embeddings of planar and bounded-treewidth graphs	2	● 2 of 2
090	Universal optimality of the triangular lattice	4	—
091	The logarithmic Brunn–Minkowski conjecture	1	● 1 of 1
092	The optimal order of convex-body covering density	2	● 1 of 2
093	Dimension-free logarithmic Sobolev inequality for subgaussian log-concave measures	1	—
094	Subpolynomial dimension reduction in L_p	1	● 1 of 1
095	Hyperbolicity cones without semidefinite lifts	3	● 1 of 3
096	The Gaussian propeller conjecture	1	● 1 of 1
097	The Euclidean Steinitz–Bergström conjecture	1	● 1 of 1
098	A negative answer to the Lang–Plaut problem	1	● 1 of 1
099	The sharp distortion of edit distance into ℓ_1	3	—
100	A counterexample to Bang's cylinder-covering bound	4	—
101	The sharp simplex conjecture for isotropic constants	1	—

Theoretical computer science · 40 families, 20 with Lean

#	Result	Papers	Lean
102	The Unique Games Conjecture and optimal approximation thresholds ▶ guide	5	● 3 of 5
103	Derandomization of logarithmic space: $L = RL = BPL$ ▶ guide	1	—
104	Quasipolynomial algorithms for mean-payoff, stochastic and parity games	4	● 1 of 4
105	The 2-to-1 Games Conjecture with perfect completeness	1	● 1 of 1
106	Hardness of coloring three-colorable graphs	1	● 1 of 1
107	Matrix multiplication with exponent at most $9/4$ ▶ guide	3	● 3 of 3
108	A cubic permanent–determinant lower bound	1	—
109	Integer multiplication below $n \log n$ ▶ guide	1	—
110	Optimal-order randomized k-server on arbitrary metrics	2	—
111	One-sample matroid prophet inequalities against an almighty adversary	1	—
112	Beyond the square-root exponent for depth-three circuits	1	● 1 of 1
113	Approximate counting and the perfect-matching entropy conjecture	2	● 1 of 2
114	Approximate counting of common integer polymatroid bases	2	● 1 of 2
115	Sampling and counting contingency tables with arbitrary margins	2	—
116	Uniform identity testing for noncommutative formulas	3	● 1 of 3
117	Uniform sparsest cut: hardness and semidefinite gaps	2	● 1 of 2
118	Unbounded bin-packing gaps and the modified integer round-up conjecture	1	● 1 of 1
119	The Courtade–Kumar and Hellinger conjectures	2	● 1 of 2
120	Almost-linear-time exact matching in general graphs	1	—
121	Almost-linear expected-time approximation of edit distance	1	—
122	Superpolynomial lower bounds and quasipolynomial reconstruction from deletion traces	3	—
124	Polynomial-time scheduling on three identical machines	1	—
125	The approximation threshold for metric k-median	2	● 1 of 2
126	Exponential semidefinite complexity of perfect matching	1	—
127	The asymptotic Gotsman–Linial conjecture	1	● 1 of 1
128	A factor-two approximation for shortest common superstring	1	● 1 of 1
129	Exponential state costs for two-way automata	2	● 2 of 2
130	Fourier transforms below $n \log n$ ▶ guide	2	● 1 of 2
131	Polynomial mixing of graph switches with prescribed degrees	1	—
132	A counterexample to the quadratic sensitivity conjecture	1	● 1 of 1
133	The complexity of Weisfeiler–Leman refinement	4	—
134	Generalized star height at most three	3	—
135	Sharp homogeneous depth-five complexity of matrix products	1	—
136	The quasilinear PCP-for-PPAD conjecture	1	—
137	One-tape time simulation in two-fifths-power space	1	—
138	Subset Sum in $O(2^{0.49n})$ time ▶ guide	2	—
139	Subpolynomial queries for log-concave sampling	1	● 1 of 1
140	Memory–sample lower bounds for noiseless Gaussian regression	6	● 5 of 6
141	The existential theory of the reals and existential–universal sentences in the counting hierarchy	1	—
142	Deterministic polynomial factorization over prime fields	1	—

Dynamical systems and ergodic theory · 12 families, 2 with Lean

#	Result	Papers	Lean
143	Uniform limit-cycle bounds in Hilbert's sixteenth problem	2	● 1 of 2
144	Banach's simple Lebesgue-spectrum problem	1	—
145	Rokhlin's multiple-mixing problem	1	—
146	Sinai's positive-entropy conjecture for the standard map	1	—
147	The near-boundary Birkhoff conjecture	2	—
148	The dimension formula for self-similar measures on the line	1	● 1 of 1
149	Permanence for weakly reversible reaction networks	2	—
150	Weak mixing of irrational triangular billiards	2	—
151	A C^1 counterexample to Shub's entropy conjecture	1	—
152	Zero entropy does not guarantee a smooth positive-volume model	1	—
153	Arithmetic classification of Bernoulli convolutions	1	—
154	Pointwise multiple ergodic averages for mixing transformations	4	—

Combinatorics · 37 families, 17 with Lean

#	Result	Papers	Lean
155	A counterexample to periodic tiling in dimension three ▶ guide	1	● 1 of 1
156	Borsuk's conjecture fails in dimension nine ▶ guide	1	● 1 of 1
157	Counterexamples to the Hadwiger and Colin de Verdière conjectures	3	—
158	The Euclidean plane cannot be colored with five colors ▶ guide	1	● 1 of 1
159	Erdős's reciprocal-sum conjecture and quasipolynomial Szemerédi bounds	1	—
160	Superexponential van der Waerden numbers	1	● 1 of 1
161	Counterexamples to Sidorenko's conjecture and the forcing conjecture	1	—
162	Counterexamples to Ryser's covering and Gyárfás's tree-cover conjectures	2	● 1 of 2
164	Hindman's finite sums and products conjecture	1	—
165	Exact crossing numbers of complete and complete bipartite graphs	2	● 2 of 2
166	The higher-dimensional Erdős distinct-distances conjecture	1	—
167	Pinned distances and a power saving for planar unit distances	2	● 1 of 2
168	Combinatorial invariance of Kazhdan–Lusztig polynomials	1	—
169	The Shareshian–Wachs e-positivity conjecture	1	—
170	Sharp logarithmic exponents for off-diagonal Ramsey numbers	2	—
171	The hypercube Ramsey conjecture	1	—
172	Classification of finite Euclidean Ramsey configurations	1	● 1 of 1
173	Seymour's second-neighborhood conjecture	1	● 1 of 1
174	Deterministic construction of strong thin spanning trees	2	● 1 of 2
175	Talagrand's conjectures and graph decompositions at expectation thresholds	3	● 2 of 3
176	The second Kahn–Kalai conjecture	1	● 1 of 1
177	Bounded-degree coboundary expanders in every dimension	1	—
178	Deterministic nonbipartite Ramanujan graphs in every fixed degree	1	—
179	The circulant Hadamard conjecture	1	● 1 of 1
180	Barnette's Hamiltonian-cycle conjecture	1	—
181	The Erdős–Gallai cycle-decomposition conjecture	1	● 1 of 1
182	Power savings for polynomial-difference-free sets	3	—
183	Power savings for planar halving lines and k-sets	1	—
184	Coloring and independence in graphs with forbidden subgraphs	2	● 1 of 2
185	Counterexamples to infinite matroid intersection and packing/covering	1	● 1 of 1
186	The Friedgut–Kalai graph and hypergraph threshold conjectures	2	—
187	Snaky in 21 Maker moves	1	—
188	The sharp constant in random triangle removal	1	—
189	Exact cycle–clique Ramsey numbers	1	—
190	Polynomial removal fails for ordered binary matrices	1	● 1 of 1
191	A power improvement in the Heilbronn triangle problem	1	—
192	A counterexample to the Gopalan–Servedio conjecture	1	—

Algebra · 18 families, 7 with Lean

#	Result	Papers	Lean
193	Serre's intersection-multiplicity conjecture	1	—
194	Lech's multiplicity conjecture	1	—
195	A counterexample to the small Cohen–Macaulay module conjecture	1	—
196	A counterexample to Kaplansky's zero-divisor conjecture ▶ guide	1	● 1 of 1
197	Nonsofic groups and group-ring counterexamples	4	● 3 of 4
198	A counterexample to the little finitistic-dimension conjecture	1	● 1 of 1
199	Counterexamples to conjectures of Auslander–Reiten, Tachikawa, and Nakayama	2	● 2 of 2
200	Eisenbud–Green–Harris and lex-plus-powers in characteristic zero	2	—
201	A counterexample to Kurosh's division-ring problem	1	—
202	The blockwise Alperin weight conjecture	1	—
203	Donovan's conjecture over fields and discrete valuation rings	2	—
204	Tensor saturation for even spin groups	1	—
205	Saxl's conjecture and universal tensor squares	2	● 1 of 2
206	Finite lattice representation: counterexamples and undecidability	2	—
207	The ℓ^1-Bass and complex Bass trace conjectures	2	● 1 of 2
208	The finite Benson–Etingof–Ostrik conjecture	1	—
209	Integral counterexamples to Gersten's conjecture	2	—
210	Foulkes' conjecture for sixth powers and quadratic stabilization	2	● 1 of 2

Probability and statistical mechanics · 29 families, 8 with Lean

#	Result	Papers	Lean
211	Geometry, diffusion, and spectra of random planar maps	7	—
212	No bigeodesics and smooth limit shapes in planar first-passage percolation	2	● 1 of 2
213	No infinite critical clusters on quasi-transitive graphs	2	● 1 of 2
214	The Benjamini–Schramm nonuniqueness conjecture	1	—
215	Massive continuum limits and exact mass asymptotics for planar $O(n)$ models	5	● 1 of 5
216	Critical logarithmic corrections and BKT scaling for the planar XY model	6	—
217	The low-temperature Sherrington–Kirkpatrick fluctuation law	2	—
218	Conformal universality for weakly interacting and disordered Ising models	6	—
219	GOE universality for random regular graphs with weak disorder	2	—
220	Directional zero–one laws and ballisticity in random environments	3	● 1 of 3
221	The Mézard–Parisi formula for diluted spin glasses	1	—
222	Perceptron free energies and microscopic jamming	4	—
223	Conformal limits of square-lattice random-cluster interfaces	6	—
224	Critical and near-critical universality for Voronoi percolation	3	—
225	Gaussian free field limits for the balanced six-vertex model	1	—
226	The double-dimer CLE_4 conjecture in the half-plane	1	—
227	The dynamical phase transition in the Sherrington–Kirkpatrick model	8	● 1 of 8
228	Continuum phase transitions for radial pair potentials	2	● 2 of 2
229	Sharp three- and four-state reconstruction thresholds	3	—
230	Exact Hausdorff measure for SLE	2	—
231	The free uniform spanning forest is a factor of IID	1	—
232	Gaussian fields and SLE interfaces for Lipschitz heights	3	—
233	The joint critical Ashkin–Teller scaling limit	1	—
234	All-temperature pressure for orthogonally invariant Ising spin glasses	1	—
235	Random-SAT thresholds, sharp variance and computability	4	—
236	The factor-of-IID threshold for free Ising states on trees	1	● 1 of 1
237	The three-quarter diameter exponent for honeycomb walks	13	—
238	Optimal logarithmic mixing of the Thorp shuffle	12	● 3 of 12
239	Sharp singularity rates for symmetric sign matrices	2	—

Mathematical logic · 6 families, 5 with Lean

#	Result	Papers	Lean
240	Shelah's eventual categoricity conjecture	2	● 1 of 2
241	Rigidity of the Turing degrees	1	● 1 of 1
242	Single-fold Diophantine representations	1	● 1 of 1
243	Choiceless counting does not capture polynomial time	2	● 1 of 2
244	The partition principle does not imply choice	1	● 1 of 1
245	The β-Barendregt–Geuvers–Klop conjecture	1	—

Group theory · 14 families, 7 with Lean

#	Result	Papers	Lean
246	Cannon's conjecture ▶ guide	1	—
247	An infinite finitely presented residually finite 2-group	2	—
248	Thompson's group F is nonamenable ▶ guide	1	● 1 of 1
249	A finitely generated counterexample to Eilenberg–Ganea	1	—
250	The Boone–Higman conjecture and higher finiteness	3	● 1 of 3
251	Amenability, unitarizability, and Ulam stability	2	● 1 of 2
252	A non-residually-finite torsion-free hyperbolic group	1	● 1 of 1
253	An infinite finitely presented simple amenable group	1	● 1 of 1
254	Classifying spaces and geometric obstructions for Artin groups	3	● 2 of 3
255	Quasi-isometric rigidity of virtually polycyclic groups	1	—
256	Howie's conjecture on equations over groups	2	—
257	A hyperbolic group with no geometric CAT(0) action	1	● 1 of 1
258	Gersten's conjecture for one-relator groups	2	—
259	A group without fixed price	1	—

Mathematical physics · 25 families, 6 with Lean

#	Result	Papers	Lean
260	Spacetime Penrose inequalities and rigidity	13	—
261	Localization and delocalization in the Anderson model	2	—
262	Sharp one-dimensional Lieb–Thirring inequalities	3	● 1 of 3
263	The ionization and generalized ionization conjectures	3	● 1 of 3
264	Strong cosmic censorship near two-ended Kerr data	3	—
265	The two-dimensional gapped area law	2	—
266	Exactly three mutually unbiased bases in dimension six	2	—
267	Positive-temperature Bose–Einstein condensation and quantum depletion	5	—
268	The spin-one Haldane gap	2	—
269	The Laughlin gap and stability under scalar disorder	2	● 1 of 2
270	Threshold and positive-energy bound states of the BFSS model	2	—
271	Bloch's law and spontaneous ferromagnetic order	4	—
272	Entanglement without secret key and the PPT-square conjecture	1	—
273	The entropy photon-number inequality	1	—
274	Moore's parity conjecture for QAC⁰	2	● 2 of 2
275	QMA-hardness of continuum Coulomb energy	2	—
276	The classical capacity of generalized amplitude damping	1	● 1 of 1
277	Threshold repetition for entangled games	1	● 1 of 1
278	Failure of Kohn–Sham ensemble representation	1	—
279	Exact quantum factoring over a fixed finite gate set	1	—
280	Strong locality for strongly rational unitary vertex operator algebras	1	—
281	QAOA optimality for the SK model	2	—
282	Scale-to-conformal enhancement in four-dimensional QFT	1	—
283	Polynomial-time, constant-error unitary synthesis from a Boolean oracle	1	—

#	Result	Papers	Lean
284	The optimal randomized–quantum query exponent	1	—

Operator algebras · 19 families, 6 with Lean

#	Result	Papers	Lean
285	Counterexamples to Baum–Connes and Kadison–Kaplansky	3	—
286	Arithmetic rigidity of lattice von Neumann algebras	2	—
287	All nonabelian free group factors are isomorphic ▶ guide	1	—
288	Kadison's similarity conjecture ▶ guide	1	—
289	The strong Kadison–Kastler conjecture	3	—
290	Connes' bicentralizer conjecture and relative bicentralizers	2	● 1 of 2
291	Toms–Winter and equivariant Jiang–Su stability	4	● 1 of 4
292	A counterexample to Kirchberg's norm-ultrapower embedding problem	1	—
293	A counterexample to the hyperinvariant-subspace problem	2	● 1 of 2
294	A counterexample to Kaplansky's quasitrace conjecture	1	● 1 of 1
295	The Kadison–Ringrose cohomology conjecture	1	—
296	The generator problem for finite factors	1	● 1 of 1
297	A ZFC counterexample to Naimark's problem	1	—
298	A counterexample to Voiculescu's free-entropy equality conjecture	1	—
299	The Kirchberg–Rørdam character criterion	1	● 1 of 1
300	The Popa–Vaes quadratic strong-operator paving conjecture	2	—
301	Classification by trace cones after Razak–Jacelon stabilization	1	—
302	The Phillips–Toms formula for minimal integer actions	2	—
303	From ordinary to strong pure infiniteness	1	—

Topology · 18 families, 1 with Lean

#	Result	Papers	Lean
304	The Hilbert–Smith conjecture in every dimension ► guide	1	—
305	Counterexamples to disk embedding and Wall's manifold conjecture	3	—
306	The purely cosmetic surgery conjecture for knots in S^3	1	—
307	Failure of rational injectivity for maximal coarse assembly	2	—
308	Smith–Toda complexes at every height with varying primes	2	—
309	The Kervaire invariant problem at the prime three	1	—
310	Quillen's conjecture in rational homology	1	—
311	Chai's invariant-ideal conjecture	1	—
312	The Grothendieck homotopy hypothesis	1	● 1 of 1
313	Finite generation for the $K(n)$-local sphere	1	—
314	The chromatic Smith fixed-point problem for finite p-groups	1	—
315	The four-dimensional Singer conjecture	1	—
316	Curtis's conjecture	1	—
317	Thomason model structures in every strict higher dimension	1	—
318	Chromatic splitting: counterexamples and filtrations	5	—
319	Counterexamples to the Hahn–Wilson conjecture at height two	1	—
320	A four-dimensional counterexample to Borel rigidity	1	—
321	A counterexample to Wall's finite $D(2)$ conjecture	1	—

Functional analysis · 11 families, 8 with Lean

#	Result	Papers	Lean
322	Tingley's sphere-isometry problem	1	● 1 of 1
323	Relative independence of the separable quotient problem	1	—
324	Lipschitz equivalence without linear isomorphism	2	● 2 of 2
325	The complete Crouzeix conjecture	2	● 2 of 2
326	The cotype–cotype conjecture under the approximation property	1	● 1 of 1
327	Markov type characterizes superreflexivity	1	● 1 of 1
328	Fixed points of nonexpansive maps in reflexive Banach spaces	1	● 1 of 1
329	A counterexample to Pietsch's metric-entropy duality conjecture	1	● 1 of 1
330	A negative answer to Kalton's Lipschitz-free approximation question	1	—
331	Asymptotic midpoint uniform convexity without asymptotically uniformly convex renorming	7	● 7 of 7
332	Metric Markov cotype two for ℓ_1	1	—

Differential geometry · 29 families, 9 with Lean

#	Result	Papers	Lean
333	Smooth isometric immersions of surfaces into $\mathbb{R}^4$	1	—
334	A smooth surface metric with no local immersion in $\mathbb{R}^3$	1	● 1 of 1
335	Gromov's integral scalar-curvature inequality	2	—
336	Spectral scalar curvature and codimension-two width	3	—
337	Cartan–Hadamard isoperimetry and CAT(0) fillings	2	—
338	Yau's uniformization conjecture	1	—
339	Katok's entropy rigidity conjecture	1	—
340	A counterexample to the nearby Lagrangian conjecture	1	—
341	Donaldson's hypersymplectic deformation conjecture	1	—
342	Donaldson's tamed-to-compatible conjecture	1	● 1 of 1
343	Sharp symplectic ball-packing criteria in higher dimensions	1	—
344	The metric Blaschke conjecture	1	—
345	Infinitely many closed geodesics on spheres and three-manifolds	1	—
346	Sharp singular-set bounds for stationary integral varifolds	2	—
347	Counterexamples to strong forms of Arnold's fixed-point conjecture	5	● 1 of 5
348	Nonnegatively curved Einstein four-manifolds and a topological gap	3	—
349	The Solomon–Yau least-volume conjecture	1	—
350	Yau's nodal conjecture: surfaces and counterexamples	3	● 2 of 3
351	Scalar curvature detects four-dimensional Ricci-flow singularities	3	—
352	Finite-time singularity of Calabi flow	1	—
353	The sharp dimension threshold for affine Bernstein rigidity	2	—
354	Isoperimetric regions in the cubic three-torus	1	—
355	Unique tangent flows at the first surface singularity	1	—
356	Gigli's characterization of Alexandrov curvature	2	● 1 of 2
357	Bi-Lipschitz coordinates at every regular RCD point	1	—
358	A three-manifold with no conjugate points and no nonpositively curved metric	1	● 1 of 1
359	Negative Kähler curvature without bounded holomorphic coordinates	2	● 1 of 2
360	Villani's convexity conjecture and regular optimal transport	2	● 2 of 2
361	Counterexamples to Yau's harmonic dimension bound	2	● 1 of 2

Partial differential equations · 16 families, 3 with Lean

#	Result	Papers	Lean
362	Global smoothness for relativistic Vlasov–Maxwell ► guide	1	● 1 of 1
363	Boltzmann nonuniqueness with exact local conservation	2	—
364	Kinetic limits and fluctuations over the Boltzmann lifespan	2	—
365	Joint metric and connection recovery from one boundary patch	3	● 1 of 3
366	The planar Mumford–Shah regularity conjecture	1	—
367	The critical dimension for the one-phase Bernoulli problem	1	—
368	The three-dimensional Ball–Evans approximation problem	2	—
369	The hot spots conjecture for simply connected planar domains	1	—
370	The subcritical Lane–Emden and Hénon–Lane–Emden conjectures	1	● 1 of 1
371	Stable blowup for a defocusing Schrödinger equation	1	—
372	Global uniqueness in smooth isotropic elasticity	1	—
373	Nonattainment in three-marginal Coulomb transport	1	—
374	Sharp one-third stability of optimal transport maps	1	—
375	De Giorgi's conjecture in dimension eight	1	—
376	Universal computation in forced Navier–Stokes flows	9	—
377	Interior $C^{1,\alpha}$ regularity for infinity-harmonic functions	1	—

Appendix B Sources

- [GitHub: openai/math](https://github.com/openai/math)
<https://github.com/openai/math>
- [Your fork: oishik1306/math](https://github.com/oishik1306/math)
<https://github.com/oishik1306/math>
- [Unite.AI: OpenAI Releases 722 Math Manuscripts From an Unreleased AI Model](https://www.unite.ai/openai-releases-722-math-manuscripts-from-an-unreleased-ai-model/)
<https://www.unite.ai/openai-releases-722-math-manuscripts-from-an-unreleased-ai-model/>
- [FourWeekMBA: OpenAI Says Internal Model Produced 722 Math Manuscripts](https://fourweekmba.com/ai-openai-says-internal-model-produced-722-math-manuscripts/)
<https://fourweekmba.com/ai-openai-says-internal-model-produced-722-math-manuscripts/>
- [OpenAI Developer Community: First look at mathematics manuscripts from an internal frontier model](https://community.openai.com/t/first-look-at-mathematics-manuscripts-from-an-internal-frontier-model-at-openai/1403886)
<https://community.openai.com/t/first-look-at-mathematics-manuscripts-from-an-internal-frontier-model-at-openai/1403886>
- [arXiv 2608.14673: A Human Audit of OpenAI's AI-Generated Mathematical Proofs](https://arxiv.org/html/2608.14673v3)
<https://arxiv.org/html/2608.14673v3>
- [The Next Web: OpenAI says its next model, Astra, has solved ten open problems](https://thenextweb.com/news/openai-astra-model-ten-math-proofs-non-sofic-groups)
<https://thenextweb.com/news/openai-astra-model-ten-math-proofs-non-sofic-groups>
- [SiliconANGLE: OpenAI's Astra solves 10 long-open math problems](https://siliconangle.com/2026/08/02/openais-astra-solves-10-long-open-math-problems-publishes-proofs/)
<https://siliconangle.com/2026/08/02/openais-astra-solves-10-long-open-math-problems-publishes-proofs/>
- [NPR: AI solved one of math's hardest problems. Humanity learned nothing \(so far\)](https://www.npr.org/2026/09/22/nx-s1-5968588/openai-navier-stokes-problem-mathematicians-learn-little)
<https://www.npr.org/2026/09/22/nx-s1-5968588/openai-navier-stokes-problem-mathematicians-learn-little>
- [The Neuron: Inside OpenAI's Navier–Stokes claim](https://theneuron.ai/news/inside-openais-navierstokes-claim-the-proof-the-ai-effort-and-the-credit-fight/)
<https://theneuron.ai/news/inside-openais-navierstokes-claim-the-proof-the-ai-effort-and-the-credit-fight/>
- [TechCrunch: OpenAI's feud with mathematicians is only escalating](https://techcrunch.com/2026/09/11/openais-feud-with-mathematicians-is-only-escalating/)
<https://techcrunch.com/2026/09/11/openais-feud-with-mathematicians-is-only-escalating/>
- [TechCrunch: OpenAI's embarrassing math \(October 2025\)](https://techcrunch.com/2025/10/19/openais-embarrassing-math/)
<https://techcrunch.com/2025/10/19/openais-embarrassing-math/>
- [Terence Tao: Announcing the Advisory Group on Mathematics and Artificial Intelligence](https://terrytao.wordpress.com/2026/09/21/advisory-group-on-mathematics-and-artificial-intelligence/)
<https://terrytao.wordpress.com/2026/09/21/advisory-group-on-mathematics-and-artificial-intelligence/>
- [Terence Tao: Hexagon](https://terrytao.wordpress.com/2026/10/06/hexagon/)
<https://terrytao.wordpress.com/2026/10/06/hexagon/>

Files in the repository used for this guide: README.md, overview.tex, CONTENTS.md, lean/formalization.yaml, lean/lakefile.lean, lean/ComparatorChallenges/*, lean/OAI/*, and reasoning_traces/irrationality-exponent-of-pi.pdf.